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$$\begin{aligned}\int \frac{\sqrt{x}-1}{\sqrt[6]{x}-1} dx &\stackrel{x=t^6}{=} \int \frac{t^3-1}{t-1} 6t^5 dt = \int \frac{6t^8-6t^5}{t-1} dt = \int 6t^7 + 6t^6 + 6t^5 dt \\ &= \frac{3}{4}t^8 + \frac{6}{7}t^7 + t^6 + c \stackrel{t^6=x}{=} \frac{3}{4}x^{\frac{4}{3}} + \frac{6}{7}x^{\frac{7}{6}} + x + c\end{aligned}$$

References